Divisibility of Shibukawa's elliptic Dedekind sum

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Abstract

We prove Conjecture 13 of Shibukawa: for coprime even a and positive odd b, the integer $bQ(a;b) = 4bs_i(a;b)$ is divisible by 4, and by 12 when $3 \nmid b$. The proof uses Shibukawa's established expression in classical Dedekind sums and the classical Meyer inversion formula. A square multiplier has positive permutation sign, which gives the factor 4. A separate elementary denominator argument gives the factor 3. We prove a more general three-term congruence for every modulus, including even moduli.

Task 1. Full solution

The domain inherited from [1] is $a \in \mathbb{Z} \setminus \{0\}$, $b \in \mathbb{Z}_{>0}$, $\gcd(a,b) = 1$, and $a + b$ odd. We retain these hypotheses in interpreting the question. Write $s_i(a;b)$ for the elliptic sum in [1] evaluated at $\tau = i = \sqrt{-1}$ and $Q(a;b)$ for its rational part. In the normalization of that paper,

$$s_\tau(a;b) = Q(a;b)\left(\frac{1}{3} - \frac{1}{6}\lambda(\tau)\right), \qquad \lambda(i) = \frac{1}{2}, \qquad 4s_i(a;b) = Q(a;b). \tag{1}$$

Theorem 1. *For every even integer $a \neq 0$ and positive odd integer b with $\gcd(a,b) = 1$,*

$$\boxed{bQ(a;b) = 4bs_i(a;b) \in 4\mathbb{Z}.}$$

If $3 \nmid b$, then $bQ(a;b) \in 12\mathbb{Z}$. In particular the latter conclusion holds for every positive modulus $b = 6m \pm 1$.

This is the full divisibility conjecture, including its stronger claim for the moduli $6m \pm 1$, in both the preprint and the published final of [1].

Define the sawtooth function by

$$((x)) = \begin{cases} x - \lfloor x \rfloor - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases} \qquad s(h;b) = \sum_{j=1}^{b-1} \left(\left(\frac{j}{b}\right)\right)\left(\left(\frac{hj}{b}\right)\right). \tag{2}$$

Here $b \geq 1$, h is any integer, and an empty sum is zero. In particular (2) gives the usual extension to noncoprime arguments.

Reduction to three classical Dedekind sums

Shibukawa's formula [1, (4.3)], obtained using Sitaramachandra Rao's Hardy-sum identity [2], is

$$Q(a;b) = 6\big(s(a;2b) + s(2a;b) - 2s(a;b)\big). \tag{3}$$

This identity is an established input. Put $a = 2c$. Directly from (2),

$$s(2c;2b) = s(c;b). \tag{4}$$

Indeed, for $1 \le j < b$ the terms indexed by j and $j+b$ have the same second sawtooth factor, whereas their first factors satisfy

$$\left(\!\left(\frac{j}{2b}\right)\!\right) + \left(\!\left(\frac{j+b}{2b}\right)\!\right) = \frac{j}{b} - \frac{1}{2} = \left(\!\left(\frac{j}{b}\right)\!\right).$$

The term indexed by b vanishes. Thus (3) becomes

$$bQ(2c;b) = 6b\big(s(c;b) + s(4c;b) - 2s(2c;b)\big). \tag{5}$$

A congruence for every unit multiplier

Theorem 2. *Let $b \ge 1$ and $h, t \in \mathbb{Z}$ satisfy $\gcd(ht, b) = 1$. Then*

$$D_b(h,t) := 6b\big(s(h;b) + s(ht^2;b) - 2s(ht;b)\big) \in 4\mathbb{Z}. \tag{6}$$

If $3 \nmid b$, then $D_b(h,t) \in 12\mathbb{Z}$.

We give a self-contained proof of the classical inversion identity used in this theorem. It is Meyer's formula [3]; see also [4, Theorem 1.2] for its attribution.

Lemma 1 (Meyer's inversion formula)**.** *Let $\gcd(h,b) = 1$, and let $I(h,b)$ be the inversion number of the permutation $\sigma_h : j \mapsto hj \pmod b$ of $\{0,\dots,b-1\}$, using least nonnegative residues. Then*

$$I(h,b) = \frac{(b-1)(b-2)}{4} - 3b\, s(h;b). \tag{7}$$

Proof. The case $b = 1$ is immediate. Suppose $b \ge 2$ and let r_j be the least nonnegative residue of hj. The values $r_1, \dots, r_{b-1}$ permute $1, \dots, b-1$. For $0 \le i < j \le b-1$,

$$r_{j-i} = r_j - r_i + b\,\mathbf{1}_{\{r_i > r_j\}}.$$

Summing this equality over all such pairs gives

$$\begin{aligned} bI(h,b) &= \sum_{k=1}^{b-1}(b-k)r_k - \sum_{k=1}^{b-1}(2k-b+1)r_k \\ &= \frac{b(2b-1)(b-1)}{2} - 3\sum_{k=1}^{b-1} k r_k. \end{aligned}$$

On the other hand, expanding (2) gives

$$s(h;b) = \frac{1}{b^2}\sum_{k=1}^{b-1} k r_k - \frac{b-1}{4}. \tag{8}$$

Eliminating the residue sum proves (7). □

Proof of Theorem 2. For $b = 1$ all sums vanish, so assume $b \geq 2$. Multiplication of residue classes gives

$$\sigma_{ht^2} = \sigma_t \circ \sigma_t \circ \sigma_h.$$

Since a square of a permutation has positive sign, $(-1)^{I(ht^2,b)} = (-1)^{I(h,b)}$. Consequently

$$I(ht^2, b) \equiv I(h, b) \pmod 2. \tag{9}$$

In (6), Meyer's identity cancels the constant term and yields

$$D_b(h, t) = 4I(ht, b) - 2I(h, b) - 2I(ht^2, b). \tag{10}$$

The last two inversion numbers have even sum by (9). Thus (10) is an integer divisible by 4.

For the additional factor 3, suppose $3 \nmid b$. For any u coprime to b, let r_j now be the residue of uj. The identity

$$2\sum_{j=1}^{b-1} j^2 = \frac{b(b-1)(2b-1)}{3}$$

and $3 \mid (b-1)(2b-1)$ imply $b \mid 2\sum j^2$. Since $r_j \equiv uj \pmod b$, also $b \mid 2\sum jr_j$. By (8),

$$2b\, s(u; b) = \frac{2}{b}\sum_{j=1}^{b-1} jr_j - \frac{b(b-1)}{2} \in \mathbb{Z}. \tag{11}$$

It follows that each of $6b\, s(h; b)$, $6b\, s(ht; b)$, and $6b\, s(ht^2; b)$ belongs to $3\mathbb{Z}$, so $3 \mid D_b(h, t)$. Together with the factor 4 already proved, this proves the theorem. □

Proof of Theorem 1. For $a = 2c$ and odd b with $\gcd(a, b) = 1$, both c and 2 are units modulo b. Take $h = c$ and $t = 2$ in Theorem 2 and use (5). Equation (1) identifies the elliptic formulation. The proof uses residue classes and therefore also covers negative a. For $b = 1$, (3)–(4) give $Q(a; 1) = 0$. □

The uniform factors cannot be increased: exact values are

$$3Q(2; 3) = 4, \qquad 5Q(4; 5) = 12, \qquad 7Q(2; 7) = 12.$$

Thus the factor 4 is sharp over all odd moduli, and the factor 12 is sharp even within each of the two classes $6m - 1$ and $6m + 1$. The contribution here is the application of permutation-sign multiplicativity to (3), with the denominator calculation (11). Formula (3) and Meyer's identity are explicitly credited prior work.

Tasks 2–5. Proposed follow-ups

Task 2. For nonzero $bQ(a; b)$, determine its exact 2-adic valuation from the Euclidean algorithm for $(a/2, b)$, and obtain a useful criterion for divisibility by 8 or higher powers of 2.

Task 3. When $3 \mid b$, determine the precise condition for $12 \mid bQ(a; b)$ and the higher 3-adic valuations of its nonzero values.

Task 4. For each fixed odd $b > 1$, determine the greatest common divisor of the values $bQ(2c; b)$ over all units c modulo b, including any exceptional moduli at which all these values vanish.

Task 5. Study denominator and divisibility bounds for the higher elliptic sums introduced by Shibukawa. Determine whether analogous permutation statistics control the rational coefficients in these Dedekind–Apostol sums.

Verification. Four exact arithmetic groups pass. An independent Hardy-sum formula checks 8,151 eligible pairs; 1,736 permutations and 61,854 unit pairs check inversions and signs directly; 140,390 triples check the general theorem, supplemented by 26 larger cases; and 3,157 signed pairs check negative arguments and period, with the boundary and sharp values. These finite checks support the proof and do not replace it. Review is by the authoring agent only.

References

[1] G. Shibukawa, *Some arithmetic properties of an elliptic Dedekind sum*, JP J. Algebra Number Theory Appl. **46**(1) (2020), 35–54, doi:10.17654/NT046010035; arXiv:1903.12200v2 (2019). Conjecture 13 is on page 52 of the final, page 13 of the preprint.

[2] R. Sitaramachandra Rao, *Dedekind and Hardy sums*, Acta Arith. **48**(4) (1987), 325–340. Identity quoted as Lemma 7 in [1].

[3] C. Meyer, *Über einige Anwendungen Dedekindscher Summen*, J. Reine Angew. Math. **198** (1957), 143–203.

[4] E. Tsukerman, *Equality of Dedekind sums mod* $\mathbb{Z}, 2\mathbb{Z}$ *and* $4\mathbb{Z}$, arXiv:1408.3225v2 (2014), Theorem 1.2.